INCH-POUND

MIL-DTL-24643/62
1 March 2007

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LSPB2SD
(INCLUDING VARIATIONS LSPB2SDW, LSPB2SDOS, LSPB2SDOSW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable is a single or double shield twisted pair intended for use with Profibus data communication systems. This cable can be manufactured in four variations: LSPB2SD (single shield with drain, non-watertight), LSPB2SDW (single shield with drain, watertight), LSPB2SDOS (double shield with drain, non-watertight), LSPB2SDOSW (double shield with drain, watertight).

REQUIREMENTS:

Qualification Required.

Construction (Waterblocked and Non-Waterblocked)

First	-	Copper conductor, solid, 22 AWG nominal.
Second	-	Polyolefin insulation. Wall thickness as necessary, in order for the completed cable to meet the specified electrical requirements.
Third	-	Two insulated conductors, twisted together with appropriate fillers to make a round cable such that they meet the electrical requirements specified below. Identification codes for pair − red and green.
Fourth	-	For watertight variants, waterblocking material shall meet the requirements of 3.3.5 and 3.3.5.3 of MIL-DTL-24643.
Fifth	-	Optional binder tape and/or binder yarn.
Sixth	-	24 AWG nominal tin-coated copper drain wire.
Seventh	-	Round, tin-coated copper braid, optimized for 01 and 02 variants.
Eighth	-	Optional binder tape or waterblocking tape.
Ninth	-	Polyolefin jacket, minimum thickness of 0.040 inch for the 01 and 02 variants. Cross-linked for 01 and 02 variants. Overall Diameter (OD): 0.360±0.030 inch for the 01 variant, 0.375±0.030 inch for the 02 variant. (This completes the 01 and 02 variants.)

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Construction Variant for Types LSPB2SDOS and LSPB2SDOSW

Tenth - For 04 variant: Waterblocking material in accordance with 3.3.5 and 3.3.5.3 of MIL-DTL-24643.

Eleventh - Optional binder tape or waterblocking tape.

Twelfth - Round, tin-coated copper braid, optimized.

Thirteenth - Optional binder tape or waterblocking tape.

Fourteenth - Cross-linked polyolefin jacket, minimum wall thickness of 0.040 inch. (This completes construction of the 03 and 04 variants.)

TABLE I. Details.

Military Part No. M24643/62	Type	Number of Pairs	Overall Diameter (inches)	
			Min.	Max.
-01	LSPB2SD-1	1	0.330	0.390
-02	LSPB2SDW-1	1	0.345	0.405
-03	LSPB2SDOS-1	1	0.440	0.500
-04	LSPB2SDOSW-1	1	0.450	0.530

EXAMINATION AND TESTS:

Basic Electricals:

DC resistance (ohms/100 m, max.)	5.45
DC resistance unbalance (percent, max.)	5
Jacket flaws	No Failure
Conductor and shield continuity	No Failure

Group A:

Visual and dimensional	No Failure
Watertightness (Types LSPB2SDW-1 & LSPB2SDOSW-1 only)	No Failure
Capacitance	
Mutual capacitance of a pair (max. @1 kHz)	3.0 nf/100 m
Characteristic impedance (0.2 MHz to 20 MHz)	150±15 ohms

TABLE II. Swept electrical parameters.

Frequency (MHz)	0.2	4.0	16	100	300
Attenuation dB/100m (max)	1.0	2.5	5.0	13.5	24.0

Group B:

Drip (95±1 °C)	Zero
Tear (pounds per inch thickness, min.) (ASTM D470)	35

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Physicals (unaged)	
Insulation	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Jacket	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Cross-linked proof test (percent, max.)	
Jacket	50
Group C:	
Physicals (aged air oven)	
Insulation (polyolefin)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (136 °C for 168 hours)	
Tensile strength (percent retained, min.)	60
Elongation (percent retained, min.)	60
Shrinkage	No Failure
Cable filler removability	No Failure
Heat distortion (percent of unaged, max.)	30
Conformance to material, construction and coverage	No Failure
Surface transfer impedance	
Milliohms per meter, max.	700
EMP response, dB, min.	60
Permanence of printing (jacket) (cycles, min.)	125
Group D:	
Flame propagation (cable)	No Failure
Qualification Inspection:	
Qualification inspection shall include basic electricals, groups A, B, C and D, plus the following:	
Aging and compatibility (cable) (95±2 °C)	No Failure
Abrasion resistance (jacket)	No Failure
Acid gas equivalent (percent, max.)	
Jacket	2

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Fillers	8
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index, max.	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, max.	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – Type A (hardness, min.)	80
Watertightness, max. leakage (cm^3)	65.5
Weathering (jacket)	No Failure

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2006-040)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – IS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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